

PROCESS FOR PREPARING DAPRODUSTAT AND COCRYSTALS THEREOF

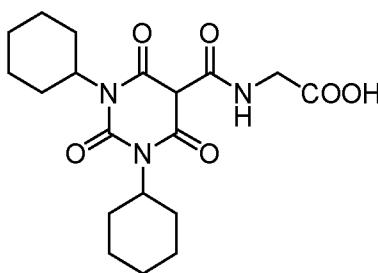
This application claims the priority of the European Patent Application EP22382718.9 filed on 26.07.2022, the European Patent Application EP22383096.9 filed on 14.11.2022, and the European Patent Application
5 EP23382239.4, filed on 14.03.2023.

FIELD OF THE INVENTION

The present invention provides crystal forms of daprodustat, which are cocrystals comprising daprodustat free acid and a daprodustat pharmaceutically acceptable metal salt, wherein the metal salt is an alkali metal
10 salt such as a sodium or a potassium salt. The invention also refers to processes for the preparation of said cocrystals of daprodustat in good yield and high purity, to pharmaceutical compositions containing them and to their use in therapy. The present invention also provides an efficient process for the preparation of daprodustat or a pharmaceutically or veterinary acceptable salt thereof in good yield and high purity suitable for industrial scale applications which involves improved conditions for the preparation of key intermediates.

BACKGROUND OF THE INVENTION

N-[(1,3-Dicyclohexylhexahydro-2,4,6-trioxypyrimidin-5-yl)carbonyl]glycine, also known as daprodustat, compound of formula (I), is an oral hypoxia-inducible factor prolyl hydroxylase inhibitor (HIF-PHI) first
20 approved in Japan under the trade name Duvroq in the form of film coated tablets in 1, 2, 4 and 6 mg dosage strengths.



(I)

Daprodustat and its tautomer *N*-[(1,3-dicyclohexyl-6-hydroxy-2,4-dioxo-1,2,3,4-tetrahydro-5-pyrimidinyl)-carbonyl]glycine were first disclosed in WO2007/150011 A2, which describes the preparation of daprodustat
25 by using *N,N'*-dicyclohexylcarbodiimide (DCC) or *N,N'*-dicyclohexylurea (DCU) as the starting material. The method involves cyclization of the starting material with malonic acid derivatives to provide 1,3-dicyclohexyl barbituric acid, following condensation with ethyl isocyanatoacetate and subsequent ester hydrolysis with aqueous sodium hydroxide in ethanol. DCC is an acute skin irritant in susceptible individuals, and due to its low melting point it is difficult to manipulate and consequently not feasible for industrial application.